F78PA Probability & Statistics A

Tutorial 2

1. For each of the following cases, determine whether the random variable is *discrete* and state the image of the random variable. Justify your answer.

   (a) A fair die is rolled once. Let X be the number showing on the die.

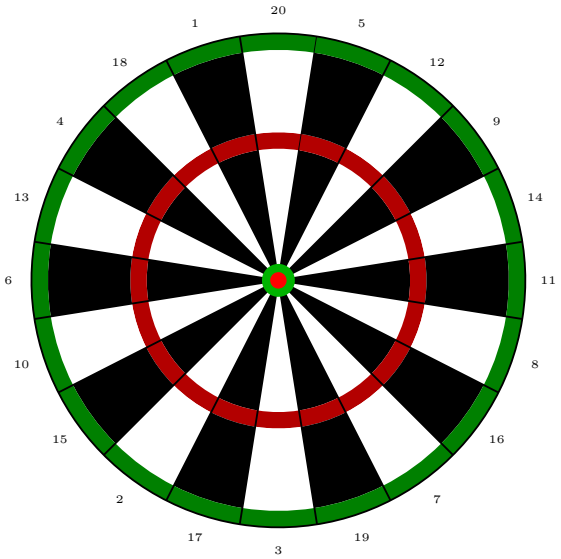
   (b) A point is chosen uniformly at random on the interval $[0, 1]$. Let $X(\omega) = \omega$.

   (c) A fair coin is tossed repeatedly until the first head occurs. Let X be the number of tosses required.

   (d) A dart is thrown at a circular board with radius 1. For simplicity assume that the dart will always hit the board. Let X be the distance from the centre of the board to the landing point of the dart.

   (e) A point is chosen uniformly at random in the interval $(0, 10)$. Let X be the integer part of the point (i.e., $X = \lfloor \omega \rfloor$).

   (f) A dart is thrown at the following board. Let X be the score given by the throw of the single dart, ignoring the possibility of doubles, triples and the bullseye.



   (g) A jar contains infinitely many marbles labelled $1, 2, 3, \ldots$, and one is drawn at random such that the probability of selecting marble n is proportional to $1/2^n$. Let X be the label of the marble drawn.

2. For each of the following discrete random variables, calculate and plot the cumulative distribution function of the random variable.

   (a) Random variable X with probability mass function $p_X(x)$ given by

x	0	1	2	3
$p_X(x)$	0.5	0.2	0.2	0.1

   (b) Random variable Y with probability mass function $p_Y(y)$ given by

y	1	2	3	4	5
$p_Y(y)$	0.05	0.05	0.1	0.2	0.6

   (c) Random variable W with probability mass function $p_W(w)$ given by

w	1	2	4	8	16
$p_W(w)$	0.2	0.2	0.2	0.2	0.2

3. A discrete random variable X has probability mass function $p_X(x)$ given by

x	0	1	2	3	4
$p_X(x)$	$a+b$	$2a+b$	$3a+b$	$4a+b$	$5a+b$

where a, b are constants. Given that $\mathbb{P}(X \geq 3) = 1/2$, find the values of a, and b.

4. Recall that X is a Binomial random variable with n trials and probability of success p if X is a discrete random variable with image $\mathrm{Im}(X) = \{0, 1, 2, \ldots, n\}$ and probability mass function

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, x \in \{0, \ldots, n\}.$$

We write $X \sim \mathrm{Bin}(n, p)$ to denote a Binomial random variable with n trials and probability of success p.

(a) For $X \sim \mathrm{Bin}(5, 0.2)$, compute $\mathbb{P}(X = 3)$.

(b) For $X \sim \mathrm{Bin}(10, 0.3)$, compute $\mathbb{P}(X \leq 2)$.

(c) For $X \sim \mathrm{Bin}(15, 0.05)$, compute $\mathbb{P}(X \geq 2)$.

(d) For $X \sim \mathrm{Bin}(8, 0.7)$, compute $\mathbb{P}(X > 6)$.

(e) For $X \sim \mathrm{Bin}(10, 0.6)$, compute $\mathbb{P}(X \geq 8.3)$.

5. A salesman sells goods by telephone. The probability that any particular call achieves a sale is 0.08, independently of all other calls. In a particular one hour period the salesman makes a total of 12 calls. Compute

i. the probability that the number of sales is 2 or fewer;

ii. the probability that the number of sales is 2 or more.

6. The quality control division of a food company is concerned about contamination of a particular product with traces of nuts. It has been found that 6% of packets of the product contain traces of nuts. A random sample of 20 packets is taken from the production line. Calculate:

i. the probability that none of the packets contains traces of nuts;

ii. the probability that more than 2 packets contain traces of nuts.

7. An electronic system contains three cooling components that operate independently. The probability of each component's failure is 0.05. The system will overheat if and only if at least two components fail. Find the probability that the system will overheat.

8. The efficacy of the influenza A vaccine is about 90%; that is, 90% of those receiving the influenza A vaccine will not contract the disease when exposed. Assume that each person's response to influenza A is independent of another person's response. How many vaccinated people must be exposed to the influenza A so that the probability of at least one person contracting the disease is at least 0.95?

9. For $X \sim \mathrm{Bin}(n, p)$, where $0 \leq p < 1$, write down an expression for $\frac{\mathbb{P}(X=r+1)}{\mathbb{P}(X=r)}$, valid for $r = 0, 1, 2, \ldots, n-1$.

Letting $x_0 = (n+1)p - 1$, show that

- For $r \leq x_0$, $\mathbb{P}(X = r+1) \geq \mathbb{P}(X = r)$
- For $r \geq x_0$, $\mathbb{P}(X = r+1) \leq \mathbb{P}(X = r)$

Explain why it follows that $\mathbb{P}(X = r)$ takes its maximal value at $r = \lfloor (n+1)p \rfloor$, where $\lfloor \cdot \rfloor$ denotes the integer part.

Exam Style Question

A company has three factories producing a particular component, in Aberdeen, Belfast and Cardiff. 30% of the components are produced in Aberdeen, 30% in Belfast, and 40% in Cardiff. Of those components produced in Aberdeen, 1% are defective; of those produced in Belfast, 3% are defective; and of those produced in Cardiff, 2% are defective.

(a) A component is chosen at random from those produced by the company. Calculate the probability that this component is defective. [2 marks]

(b) Given that a component, chosen at random, is found to be defective, calculate the probability that it was produced at the Aberdeen factory. [2 marks]

(c) A sample of 20 components is taken from all those produced by the company. Denoting by X the number of defective components in the sample, then under reasonable assumptions, X is binomially distributed.

i) What assumption(s) must we make to justify our claim that X is binomially distributed? [2 marks]

ii) Calculate the probability that more than 2 components in the sample are defective.

[3 marks]

[Total 9 marks]